Gregor Bučar

Senior Project Engineer, software and PLC automation

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SUMMARY

I build software in C# and .NET, from desktop apps and web front ends to the services behind them, mostly for GMP pharmaceutical plants where everything has to be validated and traceable. My software usually talks to real machines, so I also write PLC code when a project needs it, along with the specifications and test protocols a regulated plant requires.

EXPERIENCE

Senior Project Engineer, *Trac d.o.o., Šentjernej* — Jun 2024 – present

Software developer, student work, *Trac d.o.o., Šentjernej* — 2018 – May 2024

Trac builds automation, batch-management software and lab robots for pharmaceutical plants.

- **Batch-manufacturing platform rewrite** (.NET 10). I rebuilt the recipe engine with a state machine per step, crash recovery and a replay queue, and wrote hardware-in-the-loop tests against a real PLC, traced to the functional specification and OQ. I also wrote the approved URS, FS, SDS, HDS and OQ documents for the engine.
- **Recipe editor** (WPF). I developed the sequential-function-chart editor that process engineers use to draw batch recipes, including the condition builder. It runs at hundreds of installations with thousands of users.
- **Lab sample-preparation robot**. I wrote the module software (six Node.js/TypeScript services on one shared core, Nuxt operator screens in four languages, the audit trail, fleet deploys and central logging) and the PLC module programs on seven CODESYS controllers, with PROFINET servos, EtherNet/IP step motors, a Modbus RTU gripper and my own RS-232/RS-485 serial library. I also built the YOLO model that finds the sample tubes and the game-controller arm control, and moved the controllers to a real-time Linux kernel.
- **Installation-qualification tool** (WinUI 3, ASP.NET Core, OPC UA). I built the desktop app and the API. It tests every PLC signal live, records who tested it and the result, and has been used on five large projects.
- **Siemens and building automation** (TIA Portal, CODESYS). I wrote ISA-88 cleaning operations for an IBC washer on an S7-1500 in SCL, with my own code generators and checker, and added climate sensors and device health checks to an office building's existing CODESYS program.
- **Workload planner** (Vue, .NET). I designed the scheduling algorithm and built the app. It plans every task against people's real capacity six months ahead.
- I train customers' operators on site and work with their IT on server setup. I mentor two students, and from October 2026 I'm also on the support team.

Full-stack developer, student work, *Špica International* — Sep 2022 – May 2023

- On a time-tracking SaaS I moved whole screens from AngularJS to Angular with NgRx and built ASP.NET Core endpoints with xUnit tests for the onboarding, pricing and trial screens.

OWN PROJECTS

- **Time-tracking and invoicing SaaS**. It has a multi-tenant .NET 10 API with clean architecture, a Nuxt web app, an offline-first Expo mobile app, a tray timer and a browser extension. Tests fail the build if customers' data could mix, and it's deployed with Docker, GitHub Actions and Cloudflare. Free to use.
- **WebSCADA, now EVA** (team of five). Browser-based monitoring for fire-safety and building systems, running at several sites. I led the gateway (protocol plugins, a generic Modbus driver, keeps working through server outages) and the live floor plans, and built global linkage, the rules that turn alarms into SMS over a GSM modem with AT commands, e-mail, webhooks and device commands.
- **Home automation**. I run Home Assistant on a Raspberry Pi with my own voice assistant, a custom integration, WLED and local Tuya devices.

EDUCATION

BSc in Computer and Information Science, University of Ljubljana (FRI) — 2019 – 2023

My thesis, *Design and implementation of a decentralised digital building logbook* (2023, mentor prof. dr. Vlado Stankovski), was graded 10 out of 10.

Technical gymnasium (electrical engineering and computing), Šolski center Novo mesto 2015 – 2019

3rd prize and silver award, national ACM computing competition for secondary schools (RTK), 2018, where I placed 6th of 99 in the first group.

SKILLS AT A GLANCE

C# and .NET	●●●● Expert, since 2018	WPF with MVVM, ASP.NET Core, EF Core, SignalR
TypeScript, Vue and Nuxt	●●●○ Advanced, since 2022	Node.js services and operator web apps
Databases	●●●○ Advanced, since 2022	SQL Server and PostgreSQL in production, EF Core
Testing and deployment	●●●○ Advanced, since 2022	xUnit, Vitest, tests against a real PLC, Git, Docker, GitHub Actions
Computer vision	●●●○ Advanced, since 2023	training YOLO models, OpenCV, Python
PLC programming	●●●○ Advanced, since 2024	object-oriented Structured Text in CODESYS, SCL in TIA Portal
OPC UA and fieldbuses	●●●○ Advanced, since 2022	PROFINET, Modbus, EtherNet/IP, RS-232 and RS-485
ISA-88 batch control	●●●○ Advanced, since 2020	PLC phases and the recipe software above them
GxP validation	●●●○ Advanced, since 2022	URS, FS, SDS, OQ, installation qualification, audit trails
Agentic development	●●●○ Advanced, since 2025	Claude Code, with a spec and a plan first, tests and my review

●●●● **Expert** years of production systems with it ●●●○ **Advanced** production work on my own

LANGUAGES AND MORE

Slovenian (native), English (C2), French (A2 to B1). Driving licence B.

Project stories with diagrams and screenshots are at realgregor.github.io.